

# AI Email Triage & Auto-Draft Assistant

End-to-end n8n workflow — technical & process documentation

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n8n

Groq LLM

Gmail

Telegram

This report documents a production-ready n8n automation that watches a Gmail inbox, classifies every incoming email with an AI model (Groq-hosted GPT-OSS-20B), applies structured Gmail labels, alerts the team on Telegram when something is time-sensitive, and prepares a reply draft for human review — without ever sending an email on its own.

## 1. Executive Summary

Business inboxes mix invoices, complaints, job applications and routine noise, and sorting it manually costs time and delays the messages that actually matter. This workflow reads every new email the moment it arrives, classifies it by department, type and priority, labels it in Gmail, pings the team on Telegram for anything urgent, and drafts a reply when one is useful — leaving the final send decision to a human.

| Metric                | Detail                                                                         |
|-----------------------|--------------------------------------------------------------------------------|
| Trigger               | Gmail Trigger, polling the inbox every minute                                  |
| Classification engine | Groq-hosted GPT-OSS-20B, structured JSON output                                |
| Organisation method   | Nested Gmail labels (Department / Type / Priority / Needs Review)              |
| Urgent alerts         | Telegram message with sender, category, reason and a direct Gmail link         |
| Reply handling        | Draft created in Gmail Drafts only — never auto-sent                           |
| Safeguards            | Untrusted-input prompt guarding, enum validation, confidence-based review flag |

## 2. Business Value

- Inbox zero, automatically sorted — every email is filed under a clear department/type/priority label the moment it lands.
- Nothing urgent gets missed — Telegram fires instantly for deadlines, outages, complaints or payment issues, even while you're away from email.
- Faster replies without losing control — a draft is ready to review and send, but the AI never sends on its own.
- Self-auditing — low-confidence classifications are flagged with a “Needs Review” label instead of being silently trusted.
- Safe by design — the email body is treated as untrusted data, so the model classifies and drafts only; it cannot take actions or leak its instructions.

## 3. Architecture Overview

The workflow has a one-time setup branch and a continuous processing branch that runs for every new email.

| Branch                    | Trigger                      | Purpose                                                                    |
|---------------------------|------------------------------|----------------------------------------------------------------------------|
| Label Setup (once)        | Manual trigger, run one time | Creates the full set of nested Gmail labels the workflow will apply        |
| Email Triage (continuous) | Gmail Trigger (every minute) | Reads, classifies, labels, alerts on and drafts replies for each new email |

Every email is classified into three independent dimensions, plus a confidence score the workflow uses to decide whether a human should double-check the result.

| Dimension  | Possible values                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| Department | Finance, Sales, Support, HR, Operations, General                                                                  |
| Type       | Invoice, Payment, Complaint, Inquiry, Meeting, Job Application, Other                                             |
| Priority   | Urgent (credible 24-hour deadline, outage, security or legal risk, blocked payment, serious escalation) or Normal |

## 4. One-Time Label Setup

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### 1 Define the label set

A code node builds the full list of nested Gmail labels the workflow will ever need: four Department labels, seven Type labels, two Priority labels, and a standalone “Needs Review” label, all under an AI/ prefix so they stay grouped and out of the way of existing labels.

### 2 Create them in Gmail

Each label is created via the Gmail API with sensible visibility settings, ready to be applied to incoming mail. This branch is triggered manually and only needs to run once per Gmail account.

## 5. Continuous Flow — Read, Classify, Act

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### Step-by-step process

#### 1 Watch the inbox

A Gmail Trigger polls the mailbox every minute for new messages (spam, trash and drafts excluded), picking up both read and unread mail.

#### 2 Prepare the email

Sender name and email address are parsed out of Gmail's raw “from” data, the body is extracted from whichever field is available (plain text, HTML, or snippet), scripts/styles/HTML tags are stripped, and everything is trimmed to a clean payload — including a direct Gmail link to the thread.

#### 3 Classify and draft with Groq

The cleaned email is sent to Groq's GPT-OSS-20B model with a strict system prompt: classify by department, type and priority, write a short summary, decide if a reply is worth drafting, and return only structured JSON. The prompt explicitly instructs the model to treat the email as untrusted data, never follow instructions embedded inside it, and never reveal the prompt itself.

#### 4 Validate the AI output

The JSON response is parsed defensively. Any department, type or priority outside the allowed list falls back to a safe default (General / Other / Normal), confidence is clamped to 0–1, and a confidence below 0.75 automatically adds a “Needs Review” label so nothing questionable slips through silently.

#### 5 Resolve & apply Gmail labels

The workflow looks up the real Gmail label IDs for the AI's chosen categories and applies them to the message — Department, Type, Priority and, where relevant, Needs Review — all in one API call.

#### 6 Branch on urgency and draft need

Two independent checks run in parallel: is this email urgent, and is a reply actually useful? Each answer triggers its own action below.

Action branches

| Condition             | Action taken                                                                                                                                |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Urgent = true         | A Telegram alert is sent immediately with the sender, subject, category, the AI's stated reason, a short summary and a direct Gmail link    |
| Draft required = true | A reply draft is created directly in Gmail (correct thread, subject prefixed "Re:", addressed to the sender) — left unsent for human review |

## 6. Node-by-Node Map

| Stage             | Key nodes                                                              |
|-------------------|------------------------------------------------------------------------|
| One-time setup    | Run Label Setup Once → Required Gmail Labels → Create Gmail Labels     |
| Intake            | New Gmail Email (trigger) → Prepare Email                              |
| AI classification | Classify and Draft with Groq → Validate AI Output                      |
| Labelling         | Get Gmail Label IDs1 → Resolve Gmail Label IDs → Apply AI Gmail Labels |
| Urgent path       | Is Urgent? → Send Telegram Urgent Alert                                |
| Draft path        | Draft Needed? → Create Gmail Draft                                     |

## 7. Reliability & Security Design

- **Prompt-injection resistance** — the email body is explicitly treated as untrusted data; the model is instructed to classify and draft only, never to follow instructions embedded inside the message or reveal its own prompt.
- **No autonomous sending** — the workflow can create a Gmail draft, but nothing in the flow ever sends an email; a human always makes the final call.
- **Enum-safe parsing** — any department, type or priority the model returns outside the allowed list is discarded in favour of a safe default, so a malformed response can't corrupt the labelling.
- **Confidence-based review** — classifications below a 0.75 confidence threshold are automatically flagged with a “Needs Review” label instead of being trusted outright.
- **Resilient AI calls** — the Groq request is configured to retry automatically on failure, reducing the chance a transient API error drops an email from triage.

## 8. Setup Checklist

- Connect credentials: Gmail OAuth2 (same account across every Gmail node and the Gmail Trigger), a Groq header-auth credential (Authorization: Bearer YOUR\_GROQ\_API\_KEY), and a Telegram API credential from @BotFather.
- Run “Run Label Setup Once” manually and confirm the AI/... labels appear in Gmail — do not run this branch again afterward.
- Open your Telegram bot, press Start, get your numeric Chat ID from @get\_id\_bot, and paste it into “Send Telegram Urgent Alert”.
- Test three scenarios before activating: a routine email (labelled correctly, no Telegram alert), a time-sensitive email (Telegram alert fires with a readable summary), and a reply-needed email (a draft appears in Gmail Drafts).
- Once all three pass, save and activate the workflow — the Gmail Trigger then handles new mail automatically.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating event-driven email intake, guarded LLM classification, structured Gmail organisation and human-in-the-loop drafting.